Airline Delay Analysis

Python for Data Driven Insights

Carlos M Azofeifa
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Executive Summary

Actionable insights

- Average arrival delays differ significantly between high-traffic and low-traffic airports. Congestion and infrastructure capacity are likely key contributors to delays; classification is needed to target improvements.
- Flight cancellations are significantly higher during weather-related delays. Weather profoundly impacts operations; identify vulnerable airports for proactive disruption management.
- No statistically significant association found between cancellation status and the origin airport for the sample. Focus root cause analysis on other variables (e.g., carrier policies, network effects), not just the origin location.
- Delays from late-arriving aircraft vary significantly by month, peaking in June, July, August, and December. Implement seasonality-aware scheduling and resource optimization to manage peak travel periods effectively.
- Average carrier-related delay minutes differ significantly across carriers at sample airports. Pinpoint location-specific operational inefficiencies (crew, maintenance) within each carrier for targeted improvement.
- The number of flights delayed over 15 minutes varies significantly by season. Use temporal patterns to guide resource allocation, staffing, and schedule adjustments throughout the year.

Recommendations

- Develop an airport traffic classification model to identify and target infrastructure bottlenecks at high-congestion airports for operational improvements.
- Build a weather vulnerability predictive model to forecast potential disruptions at specific airports and proactively adjust schedules to minimize impact.
- Conduct a broader root cause analysis (beyond origin airport) to identify true drivers of cancellations (e.g., carrier policies, network effects) since airport origin is not a factor.
- Implement a seasonality-aware scheduling optimization tool to adjust buffer times and resource allocation during peak travel months (June, July, August, December).
- Create detailed carrier performance dashboards for different origins to pinpoint location-specific operational inefficiencies and drive targeted improvements.
- Develop a seasonal demand-forecasting model for significant delays (>15 min) to optimize staffing, resource availability, and schedule adjustments year-round.

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Requirements Gathering

- Business problem
  - Flight delays, cancellations, and diversions significantly harm an airline's profitability, customer trust, and brand reputation by increasing operational costs, disrupting schedules, and reducing passenger satisfaction.
  - The objective of this project is to apply data analysis to a comprehensive retrospective study of operational data from 2019 to 2023, to identify patterns and key factors contributing to flight delays, cancellations, and diversions. By uncovering operational drivers of disruption, the goal is to enable targeted improvements in scheduling, planning, and resource management that enhance reliability, reduce cost, and improve overall service quality.

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Requirements Gathering

Requirements Elaboration:

- To meet the objectives, the following set of questions need to be answered.
    1. Does the average arrival delay differ between high-traffic airports and low-traffic airports?
    2. Is the average number of flights cancelled associated with weather-related delays across airports?
    3. Is a flight's cancellation status (Yes/No) associated with the origin airport for a sample of four airports?
    4. Does the average delay caused by late-arriving aircraft differ between months of the year?
    5. Does the average carrier-related delay minutes differ across a sample of four origin airports?
    6. Does the average number of flights delayed more than 15 minutes differ across seasons?

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Data Overview

- Shape of the dataset.
  - The data set has 92477 rows and 19 columns
- Attribute data types of the dataset.
  - The attribute data types are:
    - float64(15),
    - int64(2),
    - object(2)
- Duplicate values in the dataset.
  - There are no duplicate values in the dataset

- Statistical summary of the dataset.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
year	92477.0	NaN	NaN	NaN	2020.830661	1.355159	2019.0	2020.0	2021.0	2022.0	2023.0
month	92477.0	NaN	NaN	NaN	6.227246	3.39703	1.0	3.0	6.0	9.0	12.0
carrier	92477	72	CE-011	12713	NaN	NaN	NaN	NaN	NaN	NaN	NaN
airport	92477	741	AT-72	796	NaN	NaN	NaN	NaN	NaN	NaN	NaN
arr_flights	92326.0	NaN	NaN	NaN	286.671393	910.7052	-21194.0	31.0	82.0	202.0	21873.0
arr_del15	92139.0	NaN	NaN	NaN	56.781797	164.61412	0.0	5.0	14.0	38.0	4142.0
carrier_ct	92326.0	NaN	NaN	NaN	19.492678	51.277935	0.0	1.91	5.31	15.0	1293.91
weather_ct	92326.0	NaN	NaN	NaN	2.140265	7.4867	0.0	0.0	0.27	1.75	266.42
nas_ct	92326.0	NaN	NaN	NaN	15.418681	52.716249	0.0	0.62	2.94	8.97	1485.82
security_ct	92326.0	NaN	NaN	NaN	0.175466	0.825821	0.0	0.0	0.0	0.0	58.69
late_aircraft_ct	92326.0	NaN	NaN	NaN	19.439746	64.820558	0.0	0.94	3.65	11.82	2069.07
arr_cancelled	92326.0	NaN	NaN	NaN	8.404144	53.407028	0.0	0.0	1.0	4.0	4951.0
arr_diverted	92326.0	NaN	NaN	NaN	0.749659	3.413909	0.0	0.0	0.0	0.0	154.0
arr_delay	92326.0	NaN	NaN	NaN	3839.85379	12420.268609	0.0	256.0	837.0	2462.0	438783.0
carrier_delay	92326.0	NaN	NaN	NaN	1413.232513	4498.278697	0.0	84.25	312.0	1005.0	162563.0
weather_delay	92326.0	NaN	NaN	NaN	225.316845	877.550153	0.0	0.0	14.0	138.0	27876.0
nas_delay	92326.0	NaN	NaN	NaN	745.969077	2980.517129	0.0	18.0	108.0	369.0	97283.0
security_delay	92326.0	NaN	NaN	NaN	8.28629	47.665554	0.0	0.0	0.0	0.0	3760.0
late_aircraft_delay	92326.0	NaN	NaN	NaN	1447.043336	5121.911398	0.0	34.0	231.0	871.0	227959.0

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Data Preprocessing and Feature Engineering

- Data Preprocessing
  - **Addressing inconsistent values**

From the statistics summary we can see arriving flights with negative values. This could be data errors, data processing anomalies, because the number of flights arriving at an airport can not be a negative physical count. To address this situation the abs function in python has been applied to the whole column and a following describe function allows us to confirm the changes.

  - **Rationalizing similar values**

- Carrier codes are similar however differ due to casing (i.e. cE-007, Ce-007, ce-007).

- Airport codes are also recorded in lower and upper case, that will nee to be normalized.

- These values in these columns will be converted to lower case.

***Note**: You can use more than one slide if needed*

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Data Preprocessing and Feature Engineering

- Missing value treatment

  - arr_del15 which is the Number of arriving flights delayed by 15 min or more has 0.365496% of missing values, less than 4%.

  - All other columns have 0.163284% of missing values, less than 2% so the impact on analysis is minimum.

  For the numerical values we can observe that:

  - The mean in all of the categories is significantly higher than the median showing a strong right skew
  - The maximum value in all categories is very high, indicating extreme outliers
  - In this case using the median for imputation will be better, due to mean would be distorted by outliers.

- Imputation methodology

  - The percentages of missing values is very low, however due to the presence of extreme outliers and the right skewness of the distribution, we used the median to impute missing values, preventing biases due to those extreme values.

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Data Preprocessing and Feature Engineering

Feature Engineering

- **Season**: a function to get seasonality based on the month of the year has been applied.

  In this way the seasons are established as:
  - Winter = Dec, Jan, Feb (12,1,2)
  - Spring = Mar, Apr, May (3,4,5)
  - Summer = Jun, Jul, Aug (6,7,8)
  - Fall = Sep, Oct, Nov (9,10,11)

- **Airport Traffic**: Based on the number of arriving flights, the traffic density at different airports has been categorized as:

  - 0.25 (Q1): Below this value → Low traffic.

  - 0.75 (Q3): Above this value → High traffic.

  - Values between Q1 and Q3 → Medium traffic.

- **Weather delay**: a binary feature to capture flight records with delays related to weather has been applied. After creating the flag, we now can easily identify if a particular flight record experienced a delay related to weather.
  - Delayed related to weather = 1
  - Delayed not related to weather = 0

- **Cancellation Status**: Cancellations can significantly impact airline operations and passenger experience. A binary feature was created that indicates whether a particular flight record was cancelled.

  - Cancelled flight = 1

  - Not cancelled flight = 0

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Exploratory Data Analysis

Key results from the EDA

- Univariate Analysis
  - The distribution of the years from 2019 to 2022 is relative uniform, with a slight downward trend in the proportion from 2019 to 2023, with 2023 being notably lower than previous years.
  - The distribution by month is also very uniform and matches the seasonality function results.
  - Distribution of carriers shows uneven distribution with a dominant carrier ce-011, indicating that the market is dominated by a few major players while many others have smaller operations.
  - Distribution of airport is highly uneven and skewed, suggesting significant differences in traffic volume.
  - Distribution of season across the four seasons is relatively uniform, with a difference of less than 5% between the highest and lowest proportions. Most popular seasons: Summer and Spring have the highest and nearly equal proportions of flights, each just slightly below 27%.
  - Distribution of traffic type: Uneven Distribution, High traffic type accounts for the larger proportion of the total percentage displayed, at approximately 60%. Lower traffic type accounts for a smaller proportion, at approximately 40%.
  - Distribution of Weather Delay Status: approximately 55% of the flights has a weather condition associated to the cause of the delay and 45% of the flights has no weather conditions associated to the cause of the delay

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Exploratory Data Analysis

Key results from the EDA

- Univariate Analysis

Considering the following group numerical attributes:

arr_flights	arr_del15	carrier_ct	weather_ct	nas_ct
security_ct	late_aircraft_ct	arr_cancelled	arr_diverted	arr_delay
carrier_delay	weather_delay	nas_delay	security_delay	late_aircraft_delay

The above group of attributes share the same characteristics:

- Right-Skewed Distribution: The histogram is highly right-skewed, meaning most of the data points are clustered toward the lower end of the X axis, while a long tail extends to the right due to a few very high values.
- Concentration of datapoints: A significant majority of the data falls within the lowest range of X axis. The tallest bar, which is near zero, indicates a high frequency of occurrences.
- Presence of Outliers: The long tail stretching out to the right suggests there are a few data points that are significantly different from the bulk of the data.

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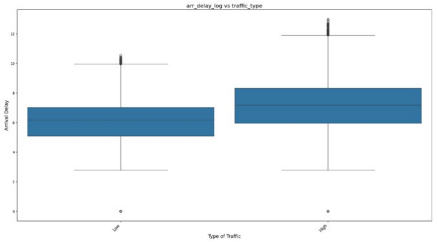
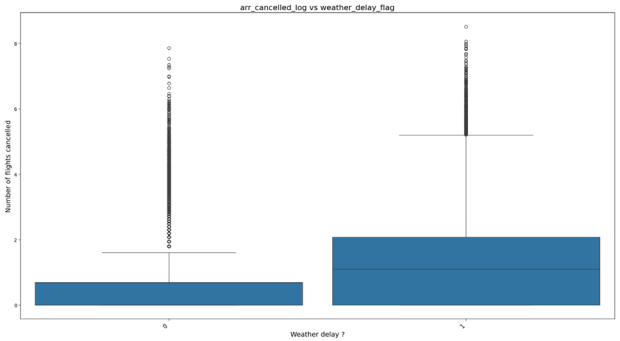
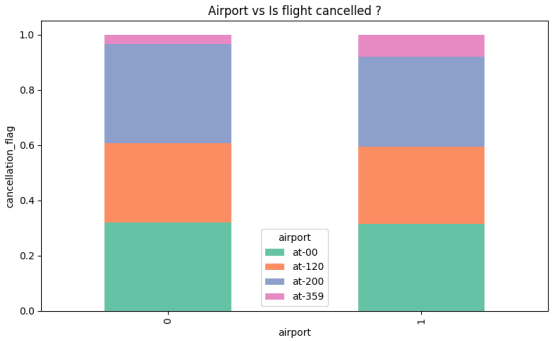
Exploratory Data Analysis

Key results from the EDA

Bivariate Analysis







arr_delay vs traffic_type:

High" traffic type consistently has higher arrival delays than the "Low" traffic type.

The range of delays is much larger for "High" traffic compared to "Low" traffic. This indicates that arrival delays are more variable and less predictable when traffic is high.

arr_cancelled vs weather_delay_flag

The median "number of cancelled flights" is much higher in the group where a weather delay was flagged (1) compared to the group with no weather delay flagged (0).

This suggests a strong positive relationship between weather delays and the incidence of cancellations. This feature is highly predictive and should be a core component of our cancellation models.

airport vs cancellation_flag

Equal Risk Distribution: The data suggests that, within this specific dataset and for these four airports randomly selected, the different airport categories contribute to cancelled and non-cancelled flights in roughly the same proportion.

There is no single airport that stands out as having a disproportionately higher rate of cancellations based solely on this comparison.

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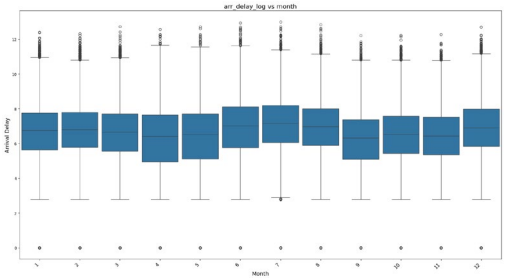
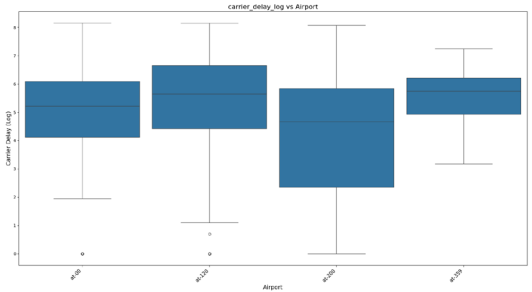
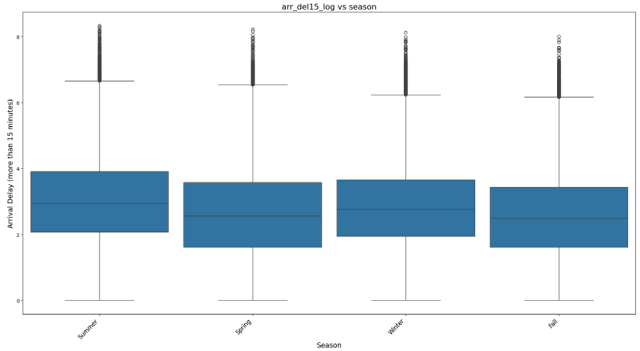
Exploratory Data Analysis

Key results from the EDA

Bivariate Analysis







arr_delay vs month

The median arrival delays are higher during the summer season (months 6, 7, and 8). Interquartile range also increases in the summer, pointing to heightened operational risk.

The presence of these clear seasonal trends validates 'month' as a useful feature for our predictive models.

carrier_delay vs airport

Medians across the airports differ. Airport at-120 consistently shows the highest typical carrier delays), while airport at-200 shows the lowest.

Variability in Risk: The varying box sizes show that operational consistency differs by location. The presence of numerous outliers across all airports highlights that severe delays are common events we must model effectively.

arr_delay_15 vs season

The data clearly shows that while median delays are low across all seasons, the variability and frequency of high-value extreme delays (outliers) are significantly more pronounced during the Summer and Winter seasons.

The "Fall" season appears to have the most consistent and lowest number of significant delays.

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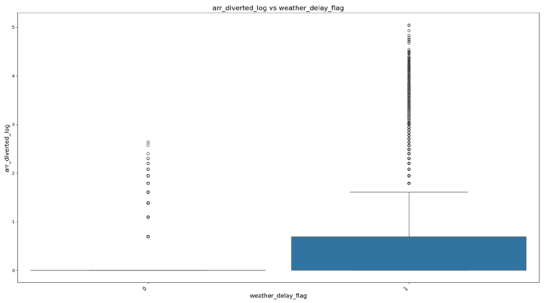
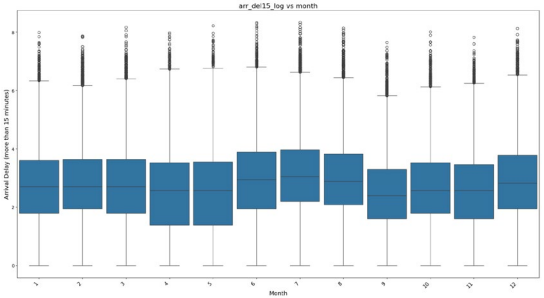
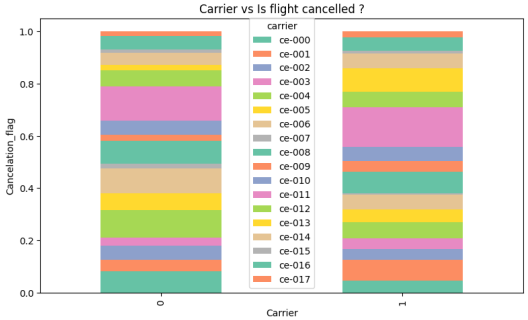
Exploratory Data Analysis

Key results from the EDA

Bivariate Analysis







arr_diverted vs weather_delay_flag

The analysis reveals a strong positive correlation, confirming weather as a primary driver of diversion events.

There is a stark difference in the distribution of diverted flights between flights that experienced a weather delay (1) and those that did not (0). The median number of diversions is significantly higher for flights with weather delays.

arr_delay_15 vs month-

Seasonal Trends: The data clearly shows that median delays (for flights already delayed 15+ minutes) are generally higher during the summer months (June, July, August) compared to the winter and fall months.

The interquartile range also increases in the summer, indicating that the typical spread of delays is wider during this time, pointing to heightened operational risk.

carrier` vs `cancellation_flag

Both non-cancelled (0) and cancelled (1) flights have almost the exact same proportional breakdown across the 18 identified carriers.

This indicates that the specific carrier alone does not seem to influence the likelihood of a flight being cancelled.

We can conclude that this single feature is unlikely to be a strong predictor for flight cancellations.

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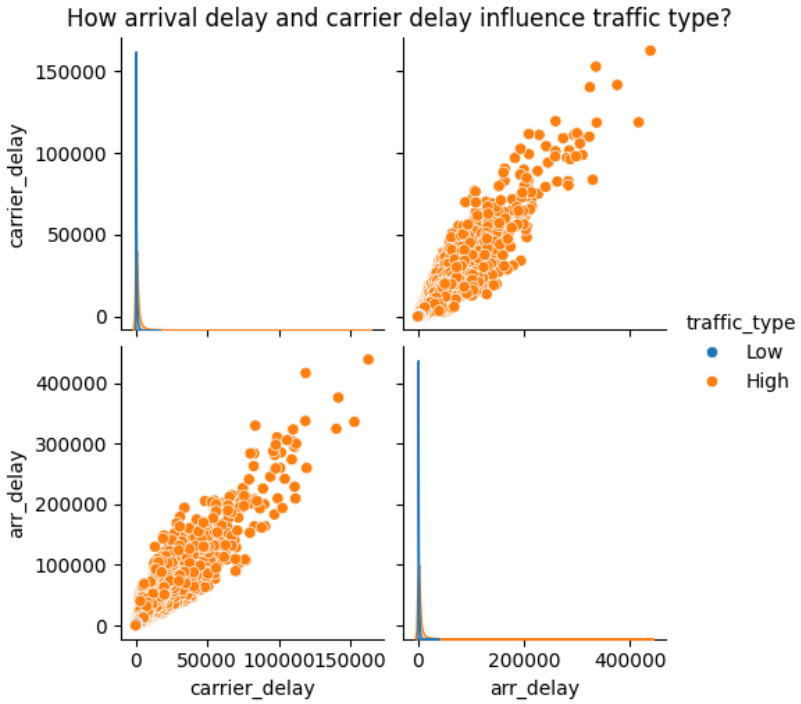
Exploratory Data Analysis

Key results from the EDA

Multivariate Analysis



Influence of `traffic_type` on `arr_delay` and `carrier_delay`

- Strong Correlation between Delays: The scatter plots show an extremely strong positive correlation between carrier_delay and arr_delay. This suggests that carrier delays are a primary driver of overall arrival delays, or vice-versa, confirming their close relationship.
- "Low" traffic type (blue) appears almost exclusively when both carrier_delay and arr_delay are zero.
- "High" traffic type (orange) appears when delays are non-zero.
- This indicates that traffic_type is likely a proxy for, or derived directly from, the delay variables. It is not an independent input feature in its current form.
- The high correlation between arr_delay and carrier_delay suggests multicollinearity. If the goal is to predict arr_delay, carrier_delay is an excellent predictor.

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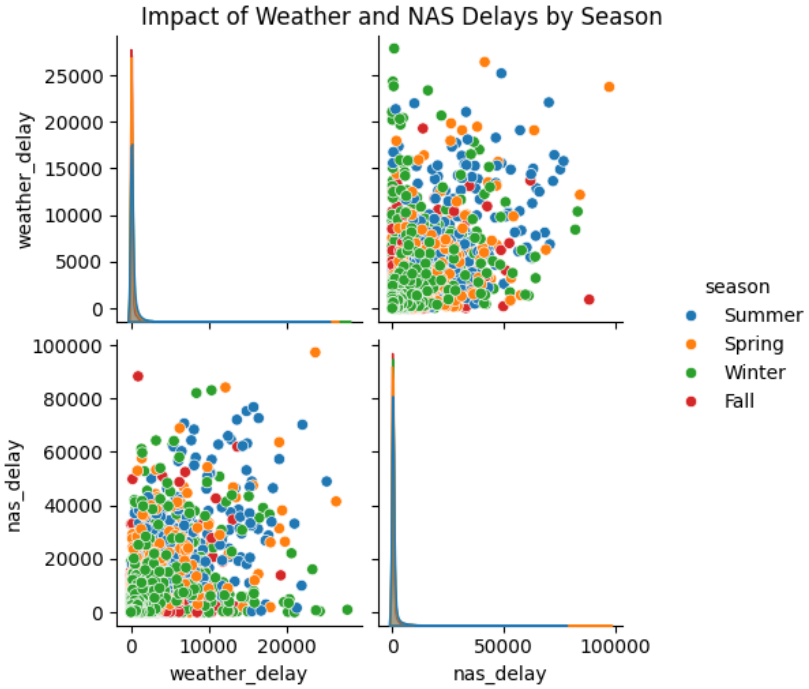
Exploratory Data Analysis

Key results from the EDA

Multivariate Analysis



Influence of `season` on `weather_delay` and `nas_delay`

- NAS delays (National Airspace System, often related to air traffic control, volume, etc.) occur much more frequently and reach higher maximum minutes than weather delays. Most data points cluster below 20,000 minutes for weather delays, while NAS delays frequently exceed 40,000 minutes and can approach 100,000 minutes.
- There appears to be a weak to moderate positive correlation between weather delays and NAS delays. This suggests that adverse weather conditions might contribute to NAS issues, but they are distinct problems, as many high-NAS-delay instances have low weather delays.
- Seasonal Impact (Hue): Winter and Spring show the most extreme NAS delays (the highest points on the y-axis in the bottom-left scatter plot). Summer is generally associated with a moderate mix of both weather and NAS delays, clustering in the middle ranges. Fall consistently shows the lowest overall delays.
- This exploratory data analysis (EDA) suggests that both weather_delay, season, and potentially other features could be strong predictors in a flight delay prediction model.

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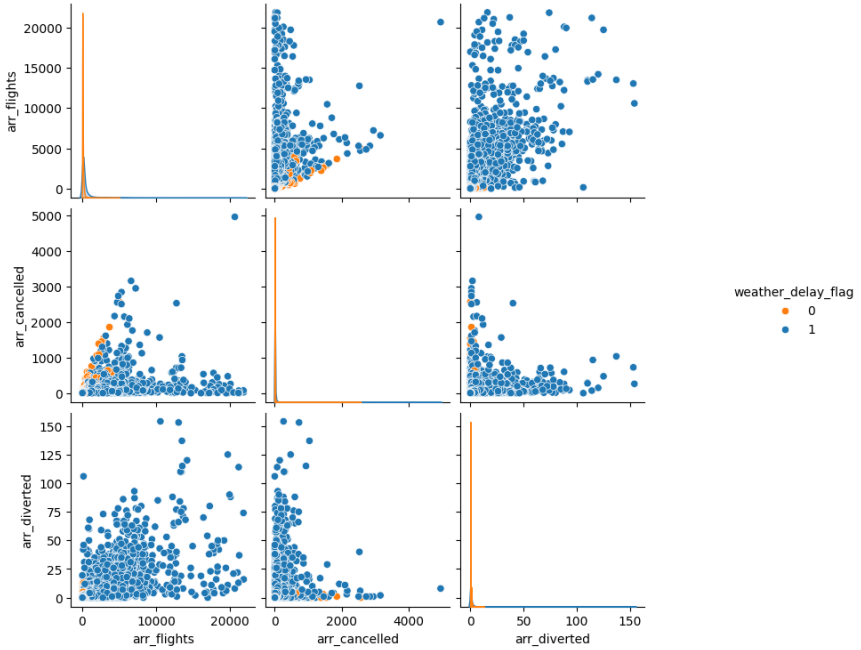
Exploratory Data Analysis

Key results from the EDA

Multivariate Analysis

How does whether a flight is cancelled due to weather or not influence the number of flights arrived, cancelled, and diverted?



Influence of `weather_delay_flag` on `arr_flights`, `arr_cancelled`, and `arr_diverted`

- Weather's Limited Role in Extreme Events: Weather delays (weather_delay_flag = 1, orange points) are associated with some cancelled and diverted flights, but the vast majority of flights (blue points, flag = 0) are still cancelled or diverted for non-weather reasons. This suggests that while weather is a factor, operational, air traffic control (ATC), or airline-specific issues are likely the larger drivers of overall flight disruptions.

Correlation:

- arr_flights and arr_cancelled: There's a positive relationship; higher flight volumes correlate with more cancellations.
- arr_flights and arr_diverted: Less clear correlation, most flights have few diversions.
- arr_cancelled and arr_diverted: Low to no significant correlation.

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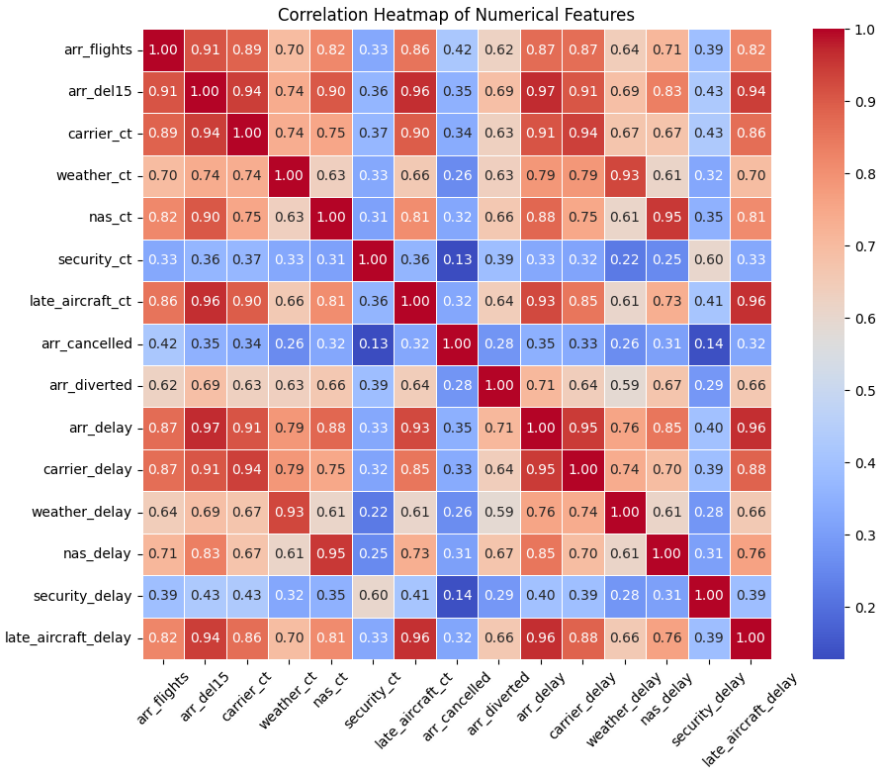
Exploratory Data Analysis

Key results from the EDA

Multivariate Analysis



Correlation Analysis

- The total arrival delay is very strongly correlated with carrier_delay, late_aircraft_delay, nas_delay, and arr_del15 (all correlations above 0.85). This confirms that operational and system-wide factors are the primary drivers of total delay.
- Weather's Significant but Secondary Role: weather_delay is strongly correlated with weather_ct (0.93), as expected. Importantly, it has moderate-to-strong correlations with the other types of delays and overall metrics (e.g., 0.74 with carrier_delay, 0.69 with arr_del15). This suggests that weather disruptions have a cascading effect across the entire system.
- Redundancy and Feature Selection: There are very high correlations (many above 0.90) among several pairs of variables. For example, late_aircraft_ct and arr_del15 have a 0.94 correlation.
- Security Delays are Unique: security_delay has low to moderate correlations with most other variables (e.g., 0.39 with arr_delay, 0.60 with security_ct), indicating that these delays are largely independent events and represent a separate operational challenge.

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Hypothesis Testing

1. Does the average arrival delay differ between high-traffic airports and low-traffic airports?

1. Null and alternative hypotheses

H_0 :The average arrival delay is the same for high-traffic airports and low-traffic airports

H_a : The average arrival delay differs between high-traffic airports and low-traffic airports.

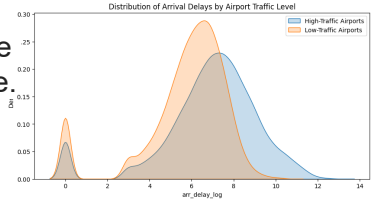
2. Choice of test and significance level

- This is a case of a two-tailed test for the significance of the difference between two independent means. Since the population standard deviations are unknown and the samples are independent, an independent samples t-test (also known as a two-sample t-test) is appropriate to test the formulated hypotheses.

- The level of significance α has been set as 0.05

3. Data preparation and assumptions check

Normality:
Low-traffic airports tend to have lower arrival delays on average.
- High-traffic airports show greater variability in delays.



Equality of variances
Levene´s test: - Statistic: 501.78, P-value: 0.0000
- Since the p-value is less than the significance level, the assumption of equal variance does not hold true.

4. p-value computation and inference

P value
T-test result - T-statistic: 84.95, P-value: 0.0000
As the p-value 0.0 is less than the level of significance (0,05), we reject the null hypothesis.

Inference
- p-value (0.0) is significantly less than α (0.05), indicating strong evidence against the null hypothesis.
- Since it's a two-tailed test, we conclude that the average arrival delay is significantly different between high-traffic and low-traffic airports.

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Hypothesis Testing

2. Is the average number of flights cancelled associated with weather-related delays across airports?

1. Null and alternative hypotheses

H_0 :The average number of flights cancelled is the same for weather-related and no weather-related delays across airports.

H_a : The average number of flights cancelled differs between weather-related and no weather-related delays across airports.

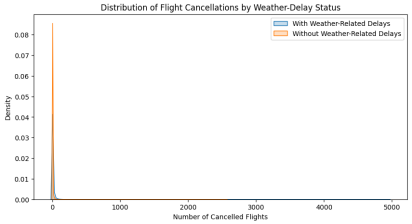
1. Choice of test and significance level

- This is a case of a two-tailed test for the significance of the difference between two independent means. Since the population standard deviations are unknown and the samples are independent, an independent samples t-test (also known as a two-sample t-test) is appropriate to test the formulated hypotheses.

- The level of significance α has been set as 0.05

3. Data preparation and assumptions check

Normality: Both distributions are right-skewed and not normal. The independent t-test assumes **normality of the distribution of the sample means**, not necessarily of the raw data.
Assumption would be satisfied due to the Central Limit Theorem



Equality of variances
Levene´s test: - Statistic: 619.41, P-value: 0.0000
- Since the p-value is less than the significance level, the assumption of equal variance does not hold true.

4. p-value computation and inference

P value
T-test result - T-statistic: 27.81, P-value: 0.0000
As the p-value 2.5806787616538193e-169 is less than the level of significance, we reject the null hypothesis.

Inference
- p-value (0.0) is significantly less than α (0.05), indicating strong evidence against the null hypothesis.

- Since it's a two-tailed test, we conclude that the average number of flight cancelled is significantly different between weather-related and no weather-related delays across airports..

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Hypothesis Testing

3. Is a flight's cancellation status (Yes/No) associated with the origin airport for a sample of four airports?

1. Null and alternative hypotheses

H_0 :Flight cancellation status is independent of the origin airport

H_a : Flight cancellation status is dependent of the origin airport

2. Choice of test and significance level

- To test the dependence between two categorical variables, we'll use a chi-square test of independence.
- The level of significance α has been set as 0.05

3. Data preparation and assumptions check

To perform the chi-squared test of independence, we need to create the contingency table

arr_cancelled	0.0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	...	21.0	22.0	29.0	30.0	36.0	37.0	49.0	70.0	104.0	106.0
airport																					
at-00	155	45	22	17	7	8	4	1	3	3	...	0	0	0	0	1	0	1	1	1	0
at-120	139	38	21	8	14	9	0	2	2	2	...	2	0	1	1	0	1	0	0	0	0
at-200	175	52	23	13	7	9	2	4	2	0	...	0	1	0	0	0	0	0	0	0	1
at-359	16	13	6	3	2	2	1	0	0	2	...	0	0	0	0	0	0	0	0	0	0

4 rows × 28 columns

4. p-value computation and inference

p-value = 0.5153955257669609

As the p-value 0.5153955257669609 is greater than the level of significance, we fail to reject the null hypothesis.

Inference

- The p-value (0.515395) is greater than α (0.05), indicating insufficient evidence to reject the null hypothesis.

- Therefore, we conclude that there is no statistically significant association between flight cancelation status and the airport of origin.

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Hypothesis Testing

4. Does the average delay caused by late-arriving aircraft differ between months of the year?

1. Null and alternative hypotheses

H_0 : The mean delay by late-arriving aircraft across all months of the year is equal.

H_a : At least one month has a mean delay by late-arriving aircraft that is different.

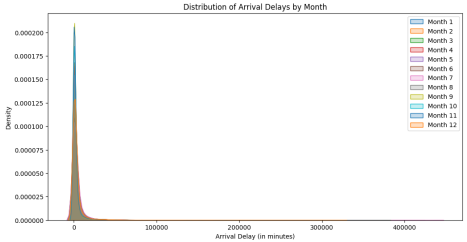
2. Choice of test and significance level

- This is a problem concerning more than two population means. We'll use a one-way ANOVA test for this case.

- The level of significance α has been set as 0.05

3. Data preparation and assumptions check

- To test normality within each group, the Shapiro-Wilk test is applied to each group.
- To test the equality of variances across groups, the Levene's test is applied.
- Since all **p-values are far less than 0.05**, we **reject the null hypothesis** for each **month of the year**.
- This provides strong statistical evidence that the **arrival delay values in each group do not follow a normal distribution**.



- The distributions for arrival delays across all months exhibit severe positive skewness (skewed to the right).

- Levene´s Test : p-value is 2.3775031739202263e-78
- Assumption of equal variances is violated across all months

4. p-value computation and inference

f_oneway() function: p-value is 5.199576695225023e-94
As the p-value 5.199576695225023e-94 is less than the level of significance, we reject the null hypothesis.

Inference

- The p-value (0) is significantly less than α (0.05), indicating strong evidence against the null hypothesis.
- Therefore, we conclude that the mean delay by late-arriving aircraft differ significantly across all months of the year.

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Hypothesis Testing

5. Does the average carrier-related delay minutes differ across origin airports?

1. Null and alternative hypotheses

H_0 : The average carrier-delay minutes across all origin airports are equal.

H_a : The average carrier-delay minutes differ across origin airports.

2. Choice of test and significance level

- This is a problem concerning more than two population means. We'll use a one-way ANOVA test for this case.

- The level of significance α has been set as 0.05

3. Data preparation and assumptions check

- To test normality within each group, the Shapiro-Wilk test is applied to each group.
- To test the equality of variances across groups, the Levene's test is applied.
- Result: since all **p-values are far less than 0.05**, we reject the null hypothesis for each **carrier**.
- This provides strong statistical evidence that the **arrival delay values in each group do not follow a normal distribution**.

Equality of variances
- Levene´s Test : p-value is 0.0039362596653060465
- Levene's test for homogeneity of variances produced a very low p-value (0), which is less than the significance level (0.05).
- We reject the null hypothesis and conclude that the assumption of equal variances is violated across all carriers.
- We can still proceed with the ANOVA, but alternative methods such as Welch's ANOVA may be more appropriate.

4. p-value computation and inference

f_oneway() function: p-value is 0.0002462019241387021

As the p-value 0.0002462019241387021 is less than the level of significance, we reject the null hypothesis.

Inference

- The p-value (0) is significantly less than α (0.05), indicating strong evidence against the null hypothesis.

- Therefore, we conclude that the average carrier delay minutes differ significantly across all carriers.

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Hypothesis Testing

6. Does the average number of flights delayed more than 15 minutes differ across seasons?

1. Null and alternative hypotheses

H_0 : The average number of flights delayed more than 15 minutes is equal across all seasons.

H_a : The average number of flights delayed more than 15 minutes differs across all seasons.

2. Choice of test and significance level

- This is a problem concerning more than two population means. We'll use a one-way ANOVA test for this case.

- The level of significance α has been set as 0.05

3. Data preparation and assumptions check

- To test normality within each group, the Shapiro-Wilk test is applied to each group.
- To test the equality of variances across groups, the Levene's test is applied.
- Result: since all **p-values are far less than 0.05**, we reject the null hypothesis for each season of the year.
- This provides strong statistical evidence that the delays 15 min or more values in each group do not follow a normal distribution.

Equality of variances
- Levene´s Test : p-value is 5.350384207132828e-55
- Since the p-value is less than the 5% significance level, we reject the null hypothesis of homogeneity of variances.
- However, since ANOVA is fairly robust to mild violations of equal variance, we can go ahead with ANOVA.

4. p-value computation and inference

f_oneway() function: p-value is 4.8700206970831525e-68

As the p-value 4.8700206970831525e-68 is less than the level of significance, we reject the null hypothesis.

Inference

- The p-value (0) is significantly less than α (0.05), indicating strong evidence against the null hypothesis.

- Therefore, we conclude that the average number of flights delayed more than 15 minutes differ significantly across all seasons.

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- Please add any other pointers (if needed)

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